

CodeFusion Hackathon



Collaborate. Innovate. Lead with AI



Team Details

- Team name:** GREENCO Hackers
- Team members:** T . Balu, G . Ranadheer, N. Nitish, P. Venkat Krishna
- Problem Statement:** Intelligent Urban Traffic Optimization and Safety System

Problem statement

- ❖ In cities, traffic congestion at signals causes long waiting times and inefficient use of road networks, making travel slow and unreliable.
- ❖ Most vehicles follow the same popular routes suggested by navigation systems, leading to overcrowding on some roads while other possible routes remain underused.
- ❖ Existing navigation systems focus mainly on shortest distance or time and do not consider **road conditions, vehicle type suitability, or safety**, which can result in heavy vehicles like trucks being guided through uneven or unsafe roads, increasing the risk of accidents.
- ❖ Poor route selection and long waiting at traffic lights cause vehicles to remain idle for longer durations, releasing harmful gases and increasing air pollution.
- ❖ This pollution negatively affects public health, causing breathing problems and discomfort for pedestrians, traffic police, roadside vendors, and daily commuters.

Addressing Problem

- ❖ Introduces a **smart traffic routing system** that dynamically distributes vehicles across **multiple routes** instead of overcrowding a single popular path.
- ❖ Considers **vehicle type and road conditions** while suggesting routes, ensuring that heavy vehicles like trucks are guided only through **safe and suitable roads**, reducing **accident risks**.
- ❖ Uses **real-time traffic data and AI-based route scoring** to select routes based not only on distance and time but also on **congestion and safety**.
- ❖ Reduces long waiting times at traffic signals by avoiding **congested paths**, thereby minimizing **vehicle idling** and improving overall **traffic flow**.
- ❖ Lowers the release of **harmful gases** by decreasing idle time and congestion, helping improve **air quality** and protect **public health**.

Social & Environmental Impact

- ❖ **Reduced traffic congestion:** Intelligent route distribution based on real-time traffic and vehicle type minimizes overcrowding on a single route and reduces traffic jams at signals.
- ❖ **Lower environmental pollution:** By reducing idle waiting time at traffic signals, the system indirectly decreases the emission of harmful gases, improving air quality around busy junctions.
- ❖ **Improved travel efficiency:** Time-optimized routing helps commuters and emergency vehicles reach destinations faster, saving fuel, time, and overall travel effort.
- ❖ **Enhanced social awareness:** The project educates users about the impact of traffic congestion on public health and encourages responsible route selection for sustainable transportation.

FEASIBILITY

- ❖ **Technical:** The project is technically feasible using web technologies and Google Maps APIs, which already provide real-time traffic data, routes, and travel time.
- ❖ **Financial:** The solution is cost-effective as it uses free or low-cost APIs and open-source tools, making it suitable for hackathons and student projects.
- ❖ **Operational:** The system is easy to operate, requiring users to only enter source, destination, travel preference, and vehicle type.
- ❖ **User-friendly:** Minimal learning is needed; the map-based interface is interactive, simple, and easy to understand for all users.
- ❖ **Scalable:** The project can be extended to city-level traffic management, emergency vehicle routing, and smart city applications.

Innovation and Uniqueness

- **Beyond shortest path routing:** While existing navigation systems mainly focus on shortest distance or time, our system intelligently distributes vehicles across multiple routes to reduce congestion instead of crowding one path.
- **Safety-aware routing:** Unlike Google Maps, which may suggest routes without considering road conditions for different vehicle types, our system avoids unsafe and uneven roads for heavy vehicles like trucks, reducing accident risks.
- **Environment-aware navigation:** Current systems do not consider pollution impact, whereas our solution actively reduces vehicle idling and harmful gas emissions at traffic signals.
- **AI-driven traffic simulation and optimization:** Existing systems react to traffic, but our system predicts and simulates traffic using AI to proactively optimize routes for time, safety, and environmental impact.

Key features of the solution

- The system intelligently distributes vehicles across **multiple routes** instead of sending all traffic through a single path, helping to **reduce congestion** and improve overall traffic flow.
- Routes are suggested based on **vehicle type and road conditions**, ensuring that heavy vehicles like trucks avoid unsafe or uneven roads and thereby **reducing accident risks**.
- The solution uses **AI-powered route optimization** to analyze traffic scenarios and select routes based on **time, congestion, safety, and environmental impact**.
- By minimizing vehicle idling at traffic signals, the system helps in **reducing harmful gas emissions**, improving **air quality**, and promoting **better public health**.

Technology Stack

Core Framework & Language :

React 19, Type Script

AI & Data :

- Google Gemini API (@google/genai)
 - gemini-3-flash-preview for smart routing & traffic simulation
 - gemini-2.5-flash for real-world location grounding

UI & Styling :

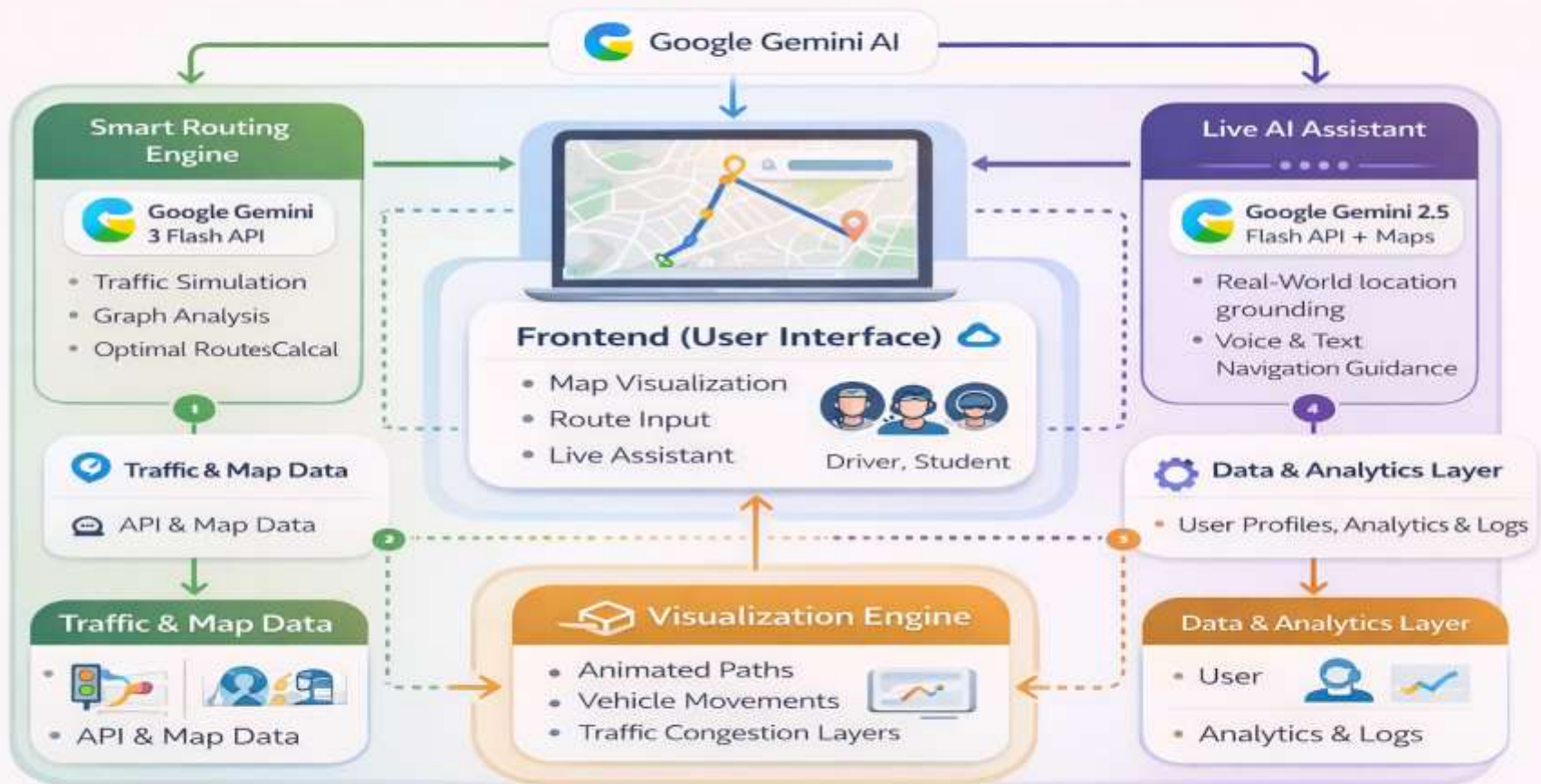
Tailwind CSS, Lucide React (Icons),
Framer Motion (Animations)

Visualization:

- Custom SVG Map System
 - Nodes, edges & traffic layers
 - Real-time path rendering & vehicle movement



AI Smart Routing System — Architecture



Project Links & Demo

Demo Video : https://drive.google.com/file/d/1iUDa2uLOZ51Ss1Ml_p0wjPKoSy6R9dyw/view?usp=sharing

MVP Link (Deployed link) : <https://nxtroute-809531510257.us-west1.run.app>

GIT Hub Repository : <https://github.com/BALU-216/NextRoute>



Thank You!